

Your grade: 100%

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Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

1. Execute the Coursera Lab Jupyter notebook for this lesson, which implements the Monte-Carlo method to approximate an integral using (1) the standard method, (2) the importance-density method, and (3) the importance-density method with weight normalization. This is the same example presented in the lesson.

1 / 1 point

What is the root-mean-squared error between the true integral value and the estimate produced by the (Monte-Carlo with importance-density and) weight-normalization method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

Note: Make sure that the code is configured with the default values of `N=10000` and `sx = 0.75` if you have changed them to experiment. Also make sure that the code executes `randn("state", 0)` before generating Gaussian random samples for the Monte-Carlo method and that it executes `rand("state", 1)` before generating uniform random samples for the (Monte-Carlo with importance-density and) weight-normalization method; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

0.0019

 **Correct**

Yes. That is correct. Interestingly, the RMSE is better than for the other two methods in this case.

2. For the same Coursera Lab Jupyter notebook, change the value of `sx` from the default of `0.75` to the new value of `1.0`. Do not modify anything else from the default conditions.

1 / 1 point

What is the root-mean-squared error between the true integral value and the estimate produced by the (Monte-Carlo with importance-density and) weight-normalization method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

0.0027

 **Correct**

Yes, that is correct.

3. For the same Coursera Lab Jupyter notebook, change the value of `N` from the default of `10000` to the new value of `25000`. Do not modify anything else from the default conditions. (Make sure that you have changed `sx` back to the default value of `0.75`, for example.)

1 / 1 point

What is the root-mean-squared error between the true integral value and the estimate produced by the (Monte-Carlo with importance-density and) weight-normalization method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

0.0034

 **Correct**

Yes, that is correct. Interestingly, using more samples gives a worse result in this case.